Supplementary methods and results

Supplementary methods

Frozen roles and populations

The gate-pilot continuation rule required at least 200 eligible enhancer-active emVars per evaluated cell or a pre-confirmation downgrade. A549 did not meet this rule in the gate analysis and was therefore fixed as descriptive before confirmation outcomes were opened. HepG2, K562, HCT116 and SK-N-SH form every confirmatory macro estimate with equal weight.

The measured, noncoding confirmation populations contained 245,897 HepG2, 245,606 K562, 151,396 HCT116 and 245,542 SK-N-SH variant-cell observations. Their fine-mapped candidate universes contained 176,396, 176,198, 119,946 and 176,100 variants, respectively. The directional enhancer-active emVar populations were 13,045, 15,809, 5,866 and 13,762. A549 counts were 100,611 measured noncoding observations, 62,319 fine-mapped candidates and 2,249 directional observations.

Deterministic policy selection

Every policy operated independently within a cell and selected exactly the requested budget where available. PIP was reduced to the maximum value across duplicated association records. Random selection was deterministic and stratified by chromosome and PIP availability. Concordance did not backfill with discordant variants. The block-capped concordance sensitivity retained at most two selected variants in each frozen hg19 1 Mb block. All ties used the same hash of the normalized hg38 variant identifier.

Bootstrap interpretation

The primary directional bootstrap rebuilt cell-specific rankings after sampling genomic blocks and is the only confirmatory inference for the concordance hypothesis. The later manuscript bootstrap held selected lists fixed, because its purpose was to attach sampling uncertainty to the already-frozen experimental allocations. The two procedures answer different questions. A conditional 10%-coverage interval from the later fixed-list risk analysis excludes zero, but it must not replace the original reranking interval, which includes zero. The manuscript therefore reports the concordance hypothesis as failed.

Cluster-robust indicator models

Each named indicator was entered alone into a logistic regression with an intercept and emVar status as the response. The sandwich covariance clustered observations by frozen hg19 1 Mb block. Absolute DNase-H3K27ac quantile difference was standardized within cell. The 20 tests comprised five indicators in each of four confirmatory cells; A549 q values were not used for confirmatory interpretation. All 20 confirmatory associations remained below FDR 0.05, but effect sizes and consistency-not significance alone-governed interpretation.

Supplementary results

Budget robustness and frozen baselines

The equal-weight magnitude-minus-AVI precision differences were 44.01 percentage points at 96 slots (1 Mb 95% CI, 35.50-52.02), 27.15 at 384 (22.93-31.22), 24.32 at 1,000 (21.87-26.82) and 12.53 at 5,000 (11.70-13.36). The 5 Mb intervals were 35.37-51.80, 22.99-31.44, 21.96-26.94 and 11.66-13.43, respectively.

Magnitude ranking also exceeded PIP and random selection at every budget. Against concordance, magnitude had 5.21 points higher precision at 96 slots (0.13-10.40), 0.91 at 384 (-1.55 to 3.27), 1.52 at 1,000 (0.14-2.84) and 2.45 at 5,000 (1.95-2.91). These hit-precision comparisons do not change the negative directional concordance result.

Genome-wide ranking discrimination

Matched magnitude exceeded AVI in both average precision and AUROC in all confirmatory cells. Average precision was 0.214 versus 0.139 in HepG2, 0.279 versus 0.137 in K562, 0.207 versus 0.113 in HCT116 and 0.171 versus 0.132 in SK-N-SH. AUROC was 0.714 versus 0.651, 0.729 versus 0.632, 0.730 versus 0.662 and 0.677 versus 0.633. PIP average precision ranged from 0.068 to 0.088 and AUROC from 0.553 to 0.598 across these cells.

Selective risk

At the prespecified 10% coverage, the frozen reranking primary contrast was +0.64 percentage points (1 Mb 95% CI, -0.04 to 1.17; 5 Mb, -0.004 to 1.12), where positive values favor magnitude. Fixed-list characterization across 1%, 5%, 10%, 25% and 50% coverage produced small, nonmonotonic contrasts, including a negative point difference at 25%. This pattern does not support a portable concordance-based abstention policy.

Rank-disagreement sensitivity

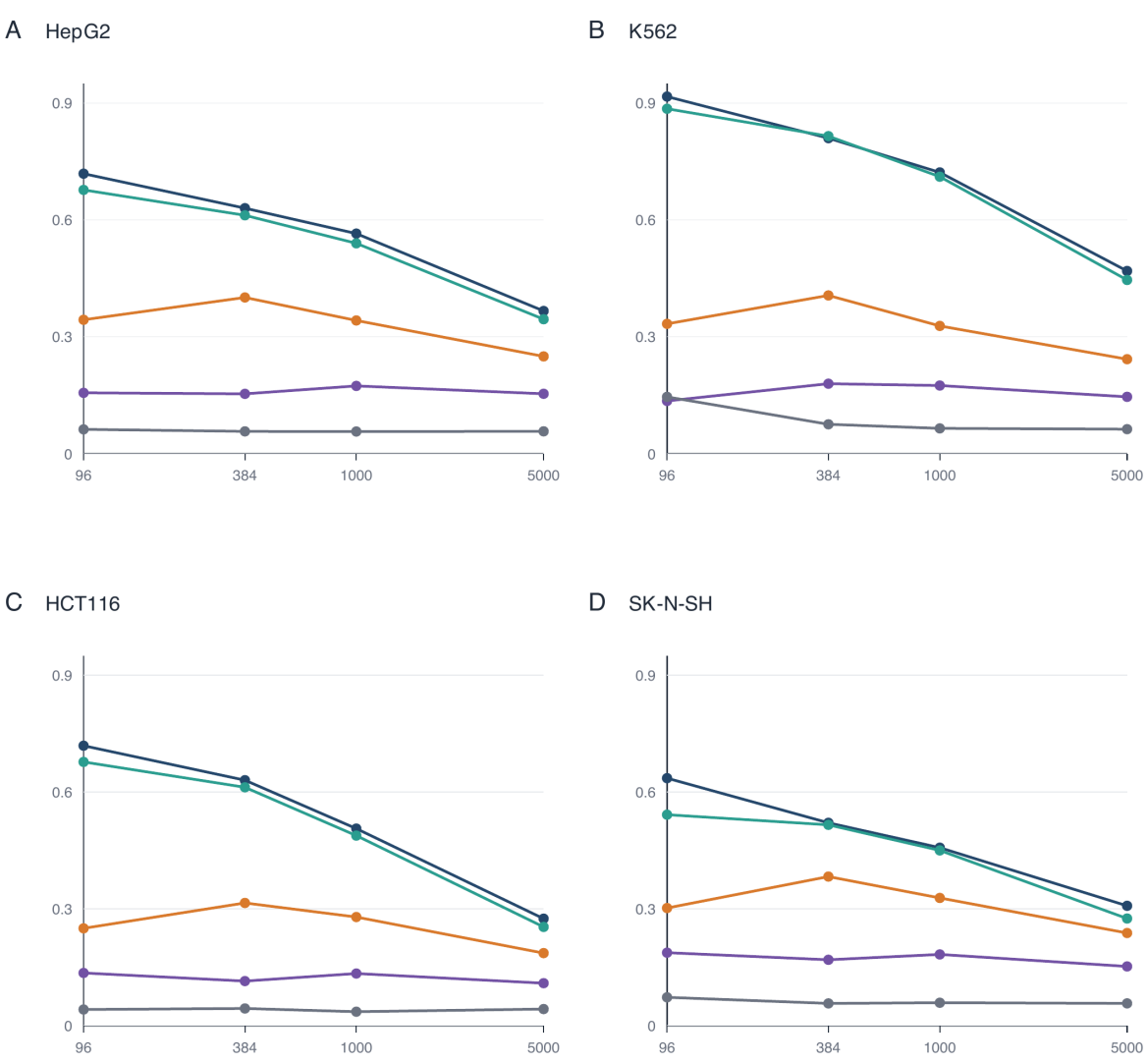
The equal-weight mean of the four cell-specific cell-matched-only versus AVI-only emVar prevalence ratios was 2.41 (1 Mb 95% CI, 2.27-2.57), with ratios of 2.10 in HepG2, 3.32 in K562, 2.70 in HCT116 and 1.54 in SK-N-SH. Using 5 Mb blocks produced the same point estimates and materially unchanged intervals. This analysis used the already-prespecified 10% threshold and did not search alternative cutoffs.

Endogenous, mechanistically adjacent sensitivity

The ADASTRA exact-cell consensus set contained 283, 662, 1,031, 213 and 93 variant-cell observations in A549, HepG2, K562, HCT116 and SK-N-SH. Magnitude-policy direction accuracy was 0.625, 0.637, 0.646, 0.587 and 0.710. Among concordance-eligible observations, accuracy was 0.677, 0.657, 0.676, 0.632 and 0.708. These small-sample results are compatible with endpoint-specific effects but do not validate MPRA direction.

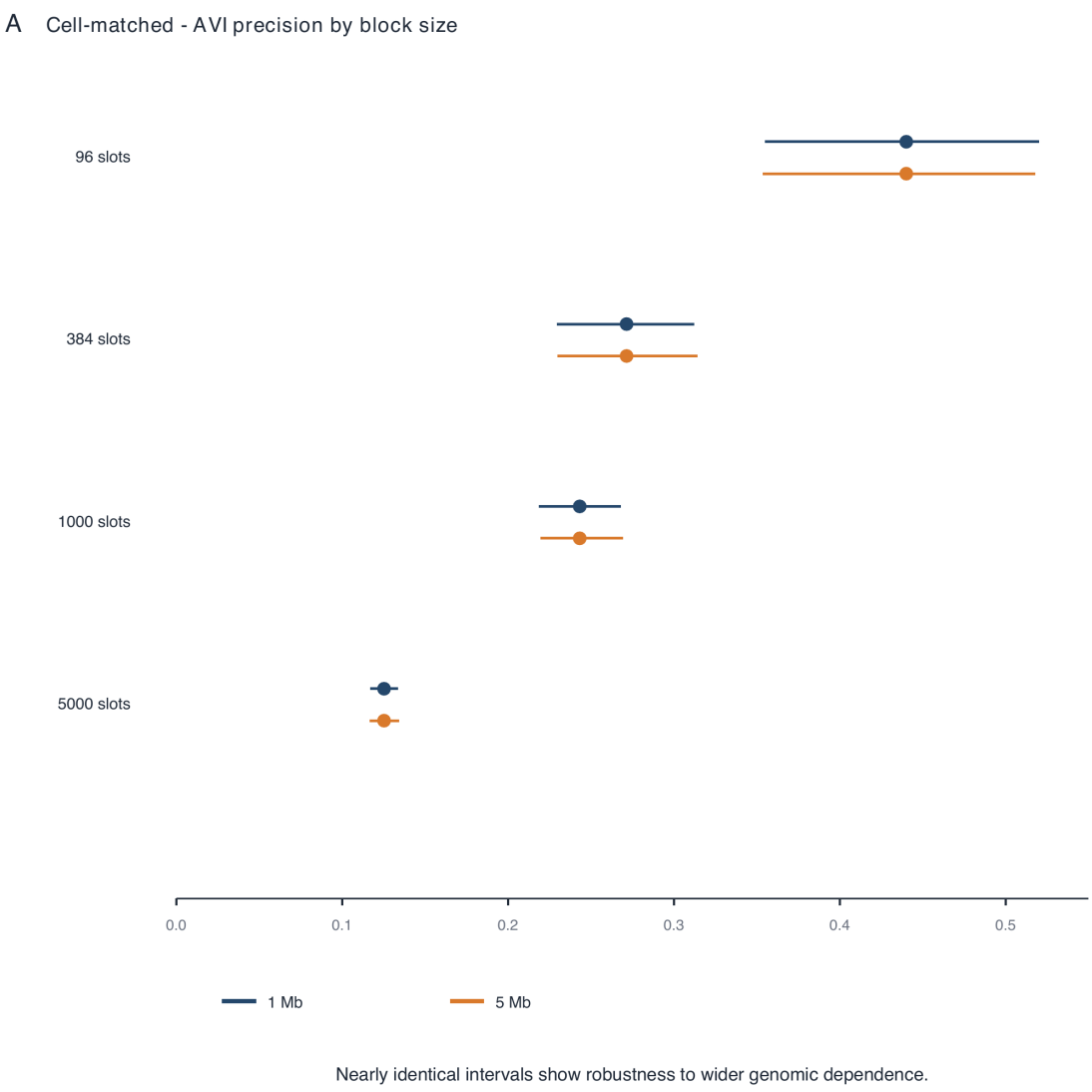
Supplementary tables

- Supplementary Table S1: all fixed-budget effect estimates and 1 Mb/5 Mb intervals (`analysis/budget_precision_contrasts.tsv`).
- Supplementary Table S2: ranking metrics (`analysis/ranking_metrics.tsv`).
- Supplementary Table S3: top-10% disagreement groups and contrasts (`analysis/avi_context_rank_groups.tsv`, `analysis/avi_context_rank_contrasts.tsv`).
- Supplementary Table S4: selective-risk intervals and contrasts (`analysis/selective_risk_intervals.tsv`, `analysis/selective_risk_contrasts.tsv`).
- Supplementary Table S5: cluster-robust uncertainty-indicator models (`analysis/uncertainty_indicator_models.tsv`).
- Supplementary Table S6: frozen exact-budget utility outcomes (`confirmation/results/budget_utility.tsv`).
- Supplementary Table S7: integrity audit and inspection receipts (`analysis/integrity_audit.json`, `analysis/inspection_receipt.json`).



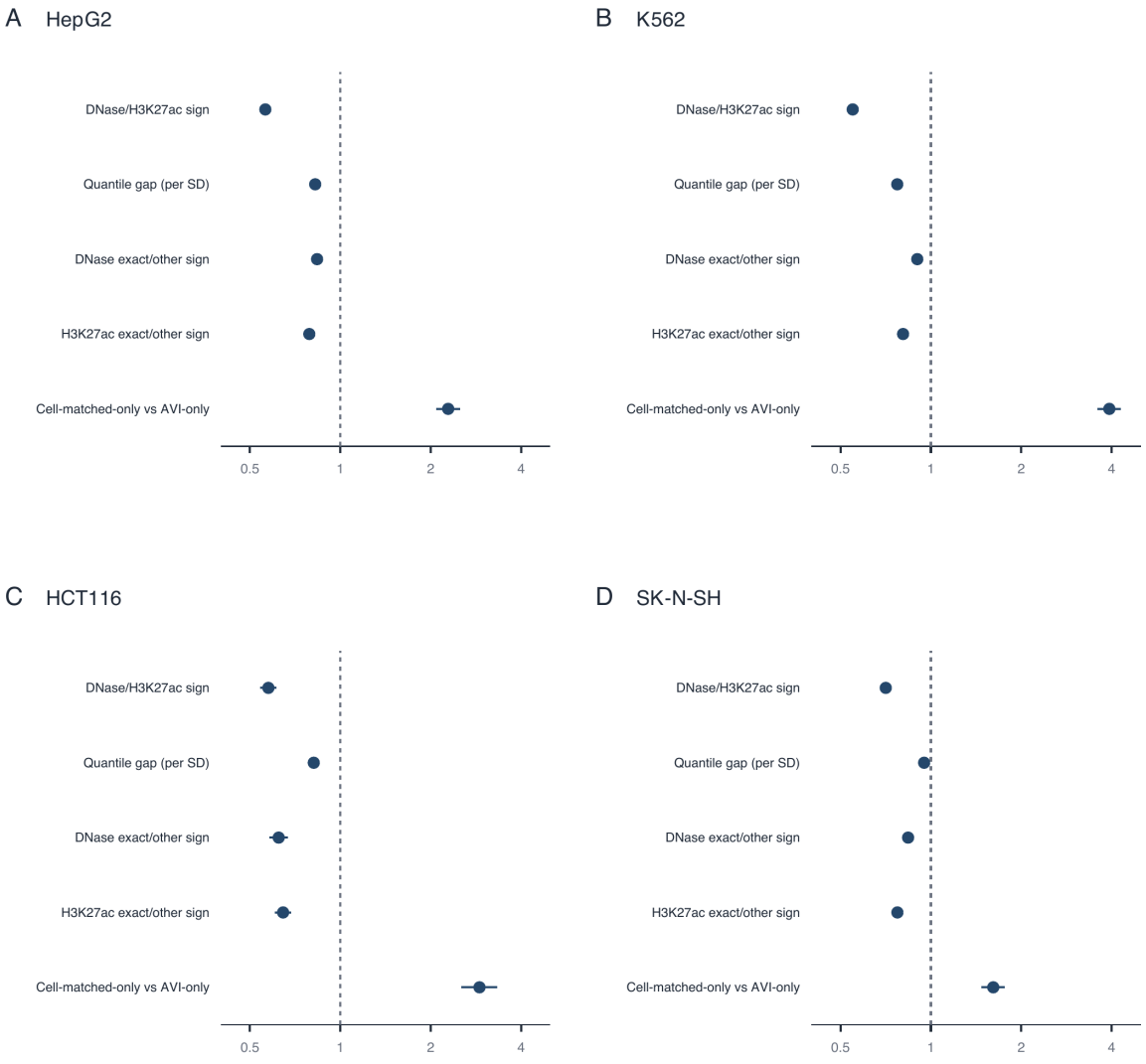
Supplementary Figure S1. Frozen policy performance by cell

EmVar precision in each confirmatory cell at the four prespecified budgets for cell-matched ranking, concordance, AVI, PIP and deterministic pseudo-random selection. Points are exact results from frozen selected lists; uncertainty for all pairwise magnitude contrasts is provided in Source Data.



Supplementary Figure S2. Dependence sensitivity

Equal-weight four-cell magnitude-minus-AVI precision differences at each frozen budget using 1 Mb and 5 Mb chromosome-stratified block-bootstrap intervals. The nearly identical intervals show that the central fixed-budget conclusion is insensitive to the wider genomic-dependence unit.



Supplementary Figure S3. Prespecified uncertainty indicators

Cell-specific odds ratios and 95% cluster-robust confidence intervals for emVar status. Binary predictors compare disagreement with agreement except for "cell-matched-only versus AVI-only," where values above one favor cell-matched-only top-10% membership. The quantile-gap odds ratio is per one population standard deviation. Dashed lines mark an odds ratio of one. Benjamini-Hochberg q values across the 20 confirmatory tests are provided in Source Data.